

What Are You Running? PuppetDB Knows.

Rich Burroughs, Daniel Parks

Puppet SRE

https://github.com/richburroughs/puppetdb_ignite

PuppetDB

Node Inventory

`web1.example.com`

`ipaddress => 10.0.0.1`
`osfamily => Debian`

Resources

`Package[bash]`

`ensure => '3.2.57'`
`web1.example.com`

How do I get it?

- **Included with Puppet Enterprise**
- **Open source users can use the puppetlabs/
puppetdb module to install and manage it**

Puppet Query Language (PQL)

Node Inventory

fact

value

certname

web1-prod.example.com

ipaddress

10.0.0.2

os.family

Debian

```
# Get information about nodes (hosts)
inventory {

    # Only nodes where the certname (FQDN) contains "prod"
    certname ~ "prod"

    # and the OS is Debian
    and facts.os.name = "Debian"
}
```



```
$ puppet query 'inventory { certname ~ "prod"
                        and facts.os.name = "Debian" }'

[
  {
    "certname": "web1-prod.example.com",
    "timestamp": "2017-03-22T19:36:20.095Z",
    "facts": {
      "ipaddress": "10.0.0.2",
      "memoryfreeinbytes": "1766612992",
      "os": {
        "name": "Debian",
        . . .
      }
    }
  }
]
```

Resources

```
class profile::database ( $password ) {  
  # This is a resource:  
  postgresql::server::db { 'myapp_database':  
    user      => 'myapp',  
    password => $password,  
  }  
}
```



```
$ puppet query 'resources { type = "Postgresql::Server::Db" }'
[
  {
    "certname": "db1-prod.example.com",
    "file": ". . . /profile/manifests/database.pp",
    "line": 3,
    "title": "myapp_database",
    "parameters": {
      "user": "myapp",
      "grant": "ALL",
      . . .
    }
  }
]
```

```
$ puppet query 'resources { type = "Postgresql::Server::Db" }'
[
  {
    "certname": "db1-prod.example.com",
    "file": ". . . /profile/manifests/database.pp",
    "line": 3,
    "title": "myapp_database",
    "parameters": {
      "user": "myapp",
      "grant": "ALL",
      . . .
    }
  }
]
```

```
$ puppet query 'resources[certname,title]
  { type = "Class" and title ~ "Role::" }'
[
  {
    "certname": "web1-prod.example.com",
    "title": "Role::Web"
  },
  {
    "certname": "db1-prod.example.com",
    "title": "Role::Db"
  },
  . . .
]
```

PuppetDB 5.0 » Puppet query

Securehttps://docs.puppet.com/puppetdb/5.0/api/query/examples-pql.html

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PuppetDBVersion: 5.0 (latest)

PuppetDB 5.0

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- + Installation
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- PQL - Puppet Query Language
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- + Query API version 4

PuppetDB 5.0 » Puppet query language (PQL) » Examples

This version of PuppetDB is not included in Puppet Enterprise. The latest version of PE includes [PuppetDB 4.4](#).

» Example Queries

- » Filtering on node names
- » Querying for active nodes
- » Basic fact filtering
- » Fact and resource filtering
- » Fact, resource and resource parameter filtering
- » Fact and resource filtering for classes
- » Fact, resource and environment filtering

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Queries in Puppet Code

```
# Find load balancer members
puppetdb_query('inventory { certname ~ "^web.*-prod" }').each |$node| {
  haproxy::balancermember { $node['facts']['fqdn']:
    listening_service => 'www',
  }
}

# Create icinga2 host objects for all nodes in inventory
puppetdb_query('inventory {}').each |$node| {
  icinga2::object::host { $node['certname']:
    'ipv4_address' => $node['facts']['ipaddress'],
  }
}
```


REST API

Rich

Requests: HTTP for Humans — x

docs.python-requests.org/en/master/

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Requests is an elegant and simple HTTP library for Python, built for human beings.



All the tools your team needs in one place. Slack: Where work happens.

ADS VIA CARBON

Requests: HTTP for Humans

Release v2.18.1. ([Installation](#))

license Apache 2.0 wheel yes python 2.6, 2.7, 3.3, 3.4, 3.5, 3.6 codecov 87% Say Thanks! 🙏

Requests is the only *Non-GMO* HTTP library for Python, safe for human consumption.

Warning: Recreational use of the Python standard library for HTTP may result in dangerous side-effects, including: security vulnerabilities, verbose code, reinventing the wheel, constantly reading documentation, depression, headaches, or even death.

Behold, the power of Requests:

```
>>> r = requests.get('https://api.github.com/user', auth=('user', 'pass'))
>>> r.status_code
200
>>> r.headers['content-type']
'application/json; charset=utf8'
>>> r.encoding
'utf-8'
>>> r.text
u'{"type": "User"... '
>>> r.json()
{'private_gists': 419, u'total_private_repos': 77, ...}
```

See [similar code, sans Requests](#).

Requests allows you to send *organic, grass-fed* HTTP/1.1 requests, without the need for manual labor. There's no need to manually add query strings to your URLs, or to form-encode your

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```
>>> import requests
>>> url = "http://httpbin.org/ip"
>>> r = requests.get(url)
>>> print(r.text)
{
  "origin": "10.0.0.1"
}
```

```
def get_nodes():  
    nodes = []  
    url = "http://localhost:8080/pdb/query/v4/nodes"  
    r = requests.get(url)
```

```
def get_nodes():  
    nodes = []  
    url = "http://localhost:8080/pdb/query/v4/nodes"  
    r = requests.get(url)  
    response = json.loads(r.text)  
    for i in response:  
        nodes.append(i['certname'])  
    return nodes
```



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Slides on GitHub
https://github.com/richburroughs/puppetdb_ignite